

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2022-23

Paper: GLB-501 — Physical and Structural Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 052735

*Instructions: Answer **five** questions in all, selecting at least **two** questions from each of Sections A and B. Section-C is compulsory. All questions carry equal marks.*

SECTION — A

Question 1. Describe the erosional and depositional features formed by glaciers. [14]

Question 2. Briefly discuss the following: [14 = 4 + 5 + 5]

- (a) Landscape Plain
- (b) Wegener's Continental Drift Theory
- (c) Drainage patterns

Question 3. Write a short note on the following: [14 = 4 + 5 + 5]

- (a) Mid-oceanic ridges with suitable examples
- (b) Isostasy, proposed by Pratt and Airy
- (c) Define landslide and discuss the mitigation of landslides

SECTION — B

Question 4. Define Joint and different joint terminologies with a neat diagram. Discuss the classification of joints based on spatial relationship and geometry with neat diagrams. [14]

Question 5. What are the fundamental possibilities of stress regimes and stress orientations that produce three types of faults? Discuss with neat diagrams. [14]

Question 6. Describe the difference between the following: [14 = 4 + 4 + 4 + 2]

- (a) Window and Klippe [4]
- (b) Listric and Detachment fault [4]
- (c) Reclined and Recumbent folds [4]
- (d) S-fabrics and L-fabrics [2]

Question 7. Discuss different parameters required to understand Ramsay's geometrical classification of folds with a neat diagram. [14]

SECTION — C

Question 8. Discuss each of the following in two or three sentences. Give suitable diagrams wherever necessary: [14 = 3 + 3 + 3 + 3 + 2]

- (a) Spherical and stereographic projection of a dipping plane [3]
- (b) Nonconformity [3]
- (c) Onshore, Backshore, Foreshore, and Offshore [3]
- (d) Epirogeny and orogeny [3]
- (e) Strain distribution on a profile of a buckling fold [2]